Assignment 3

Ulric Aird
Department of Computer & Technology
Introduction to Artificial Intelligence (COMP 300)
Professor Senthilkumar Somasundaram
November 16th, 2026

1. Finding the Goal State from S to G

A. Breadth First Search

Node Expanded	Queue	Closed List
-	(S)	-
S	(A, B, C)	{ S }
A	(B, C, D, E)	{ S, A }
B	(C, D, E, F, G)	{ S, A, B }
C	(D, E, F, G, H)	{ S, A, B, C }
D	(E, F, G, H)	{ S, A, B, C, D }
E	(F, G, H)	{ S, A, B, C, D, E }
F	(G, H)	{ S, A, B, C, D, E, F }
G	(H)	{ S, A, B, C, D, E, F, G }

Path: G → B → S

B. Uniform Cost Search

Node Expanded	Priority Queue (Cost, Path)	Closed List
-	(0, S)	-
S	(3, AS) (5, CS) (6, BS)	{ S }
A	(5, CS) (6, BS) (11, EAS) (12, DAS)	{ S, A }
C	(6, BS) (11, EAS) (12, DAS) (12, HCS)	{ S, A, C }
B	(11, EAS) (12, DAS) (12, HCS) (18, FBS) (20, GBS)	{ S, A, C, B }
E	(12, DAS) (12, HCS) (18, FBS) (20, GBS)	{ S, A, C, B, E }
D	(12, HCS) (18, FBS) (20, GBS)	{ S, A, C, B, E, D }
H	(17, IHS) (18, JHS) (18, FBS) (20, GBS)	{ S, A, C, B, E, D, H }
I	(18, JHS) (18, FBS) (18, KIHS) (19, MIHS) (20, GBS)	{ S, A, C, B, E, D, H, I }
J	(18, FBS) (18, KIHS) (19, MIHS) (20, GBS)	{ S, A, C, B, E, D, H, I, J }

Node Expanded	Priority Queue (Cost, Path)	Closed List
F	(18, KIHS) (19, MIHS) (20, GBS)	{ S, A, C, B, E, D, H, I, J, F }
K	(19, MIHS) (20, GBS)	{ S, A, C, B, E, D, H, I, J, F, K }
M	(20, GBS)	{ S, A, C, B, E, D, H, I, J, F, K, M }
G	Goal reached	{ S, A, C, B, E, D, H, I, J, F, K, M, G }

Path: S → B → G

Cost: 6 + 14 = 20

C. Depth-First Search

Node Expanded	Stack	Closed List
-	(S)	-
S	(A, B, C)	{ S }
A	(D, E, B, C)	{ S, A }
D	(E, B, C)	{ S, A, D }
E	(B, C)	{ S, A, D, E }
B	(F, G, C)	{ S, A, D, E, B }
F	(G, C)	{ S, A, D, E, B, F }
G	(C)	{ S, A, D, E, B, F, G }

Path: S → B → G

2. **Finding Goal State From M to R**

A. Breadth First Search

Node Expanded	Queue	Closed List
-	(M)	-
M	(A, D, E)	{ M }
A	(D, E, B, R)	{ M, A }
D	(E, B, R)	{ M, A, D }
E	(B, R)	{ M, A, D, E }
B	(R, C)	{ M, A, D, E, B }
R	(C)	{ M, A, D, E, B, R }

Path: M → A → R

B. Depth First Search

Node Expanded	Stack	Closed List
-	(M)	-
M	(A, D, E)	{ M }
A	(B, R, D, E)	{ M, A }
B	(C, D, E, R)	{ M, A, B }
C	(D, E, R)	{ M, A, B, C }
D	(E, R)	{ M, A, B, C, D }
E	(R)	{ M, A, B, C, D, E }
R	Goal reached	{ M, A, B, C, D, E, R }

Path: M → A → R

C. Uniform Cost Search

Node Expanded	Priority Queue	Closed List
-	(0, M)	-
M	(1, EM) (4, DM) (5, AM)	{ M }
E	(4, DM) (5, AM) (41, REM)	{ M, E }
D	(5, AM) (9, EDM) (12, BDM) (25, RDM) (41, REM)	{ M, E, D }
A	(7, BAM) (9, EDM) (12, BDM) (25, RDM) (35, RAM)	{ M, E, D, A }
B	(9, EDM) (12, BDM) (19, CBAM) (25, RDM) (35, RAM)	{ M, E, D, A, B }
R	Goal reached	{ M, E, D, A, B, R }

Path: M → D → R

Cost: 4 + 21 = 25

D. Greedy Best-First Search

Node Expanded	Priority Queue	Closed List
-	(25, M)	-
M	(2, EM) (35, DM) (40, AM)	{ M }
E	(0, REM) 35, DM) (40, AM)	{ M, E }
R	Goal reached	{ M, E, R }

Path: M → E → R

Cost 1 + 40 = 41

E. A* Search

Node Expanded	Priority Queue	Closed List
-	(25, M)	-
M	(3, EM) (39, DM) (45, AM)	{ M }
E	(39, DM) (41, REM) (45, AM)	{ M, E }
D	(25, RDM) (37, BDM) (41, REM) (45, AM)	{ M, E, D }
R	Goal reached	{ M, E, D, R }